



Figure 1: *Diagram of our model in the train and sampling phases. Solid lines represent deterministic mappings and dashed lines represent sampling. The conditioning variable enters the network in base density $p(\mathbf{z}|\mathbf{x})$ and the bijective mappings $f(\mathbf{y}, \mathbf{x})$.*

Conditional Prior

$$p(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \mu(\mathbf{x}), \sigma^2(\mathbf{x}))$$

Conditional Split Prior

$$p(\mathbf{z}_1|\mathbf{z}_0, \mathbf{x}) = \mathcal{N}(\mathbf{z}_1; \mu(\mathbf{z}_0, \mathbf{x}), \sigma^2(\mathbf{z}_0, \mathbf{x}))$$

Conditional Coupling

$$\mathbf{y}_0 = s(\mathbf{z}_1, \mathbf{x}) \cdot \mathbf{z}_0 + t(\mathbf{z}_1, \mathbf{x}); \quad \mathbf{y}_1 = \mathbf{z}_1$$